

Course: Health Economics

An Econometric Analysis of the Determinants of Subjective Well-Being

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I- Introduction

Happiness, or subjective well-being, is a central indicator of individual and collective quality of life. Unlike objective measures of living standards such as income, happiness reflects each individual's own perception of their life. This subjective dimension makes it a particularly complex object of study, but an essential one for informing public policy and understanding social dynamics.

At a time when socio-economic inequalities persist and well-being occupies a growing place in public debate, it is crucial to identify the factors that shape the probability of being happy. Subjective well-being is often used as a measure of positive mental health, alongside depression and anxiety scales. Individuals who report high levels of happiness show greater psychological resilience to negative events, adopt healthier behaviors (physical activity, diet, sleep), and report fewer depressive symptoms and less stress (Steptoe et al., 2015).

According to the definition of the World Health Organization (WHO, 1948) [1], "Health is a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity."

Studying the determinants of individual well-being touches on several dimensions of health. In terms of mental health, well-being is an indicator of emotional stability and of the absence of depression or anxiety. In terms of physical health, it is associated with better immunity, longevity and health behaviors. It also captures the quality of relationships and social support (social health) and complements objective indicators such as mortality and morbidity.

II- Research Question

To what extent do individuals' socio-economic and demographic characteristics influence their probability of being happy?

This central question can be broken down into several more specific questions:

- Does gender have a significant effect on reported happiness?
- Do education and income play a protective role?
- Do health behaviors (smoking, alcohol consumption) affect well-being?
- Do financial difficulties significantly reduce the probability of being happy?

1. WHO Constitution

III- Objectives of the Study

1. Identify the statistically significant determinants of happiness
2. Quantify the effect of each variable on the probability of being happy
3. Compare alternative econometric specifications (Logit vs. Probit)
4. Assess the robustness of the results

After a brief literature review, we present the data and descriptive statistics. We then describe the econometric methodology, followed by the results and robustness checks. The conclusion summarizes the main findings.

IV- Literature Review

The economic literature on happiness has grown considerably since Easterlin's (1973) pioneering work, which highlighted the *happiness paradox*: beyond a certain income threshold, increases in wealth do not lead to proportional gains in subjective well-being.

Empirical studies converge on several robust findings regarding socio-economic factors. Income has a positive but diminishing effect on happiness (Kahneman & Deaton, 2010). Education affects well-being indirectly through employment and social status (Cuñado & de Gracia, 2012). Unemployment and financial instability have large negative effects, often larger than the loss of income alone (Clark & Oswald, 1994).

Evidence on the effect of sex is mixed. Some studies find that women report slightly higher levels of happiness (Inglehart, 2002), while others find no significant gap once socio-economic variables are controlled for.

Regarding health behaviors, the literature shows that smoking is generally associated with lower well-being. Using international data, Weinhold and Chaloupka (2017) show that smokers report lower life satisfaction than non-smokers, even after controlling for socio-economic variables. These behaviors may be both a cause and a consequence of poor well-being.

Logit and Probit models are commonly used to analyze the determinants of happiness when the dependent variable is dichotomized or ordinal (Ferrer-i-Carbonell & Frijters, 2004). These approaches model the probability of belonging to a given well-being category as a function of qualitative or quantitative explanatory variables.

Hypotheses :

Drawing on this literature, we formulate three main hypotheses regarding the economic and behavioral factors likely to influence individual well-being.

H1: Higher income increases the probability of being happy.

H2: Financial difficulties significantly reduce well-being.

H3: Smokers report lower levels of happiness than non-smokers.

V- Data

The data come from a survey conducted by the University of Angers among adults living in France. The questionnaire collects a wide range of information on respondents' socio-demographic characteristics, economic situation, working conditions and health behaviors.

The initial survey covered 1,220 individuals aged 18 and over. Following the data preparation process, the final dataset used for the econometric analysis consisted of 1,049 observations, with no missing values in the variables retained for the analysis.

A key advantage of this dataset is that it combines economic and social information (income, education, employment, housing) with health indicators (smoking, alcohol consumption, vaccination, physical working conditions). It therefore provides a relevant empirical basis for studying the links between well-being and socio-economic characteristics.

V-.1 Dependent variable (bienetre)

Subjective well-being (bienetre):

"Overall, how happy are you with your life?"

Coding: scale from 1 to 10, where 1 = not happy at all and 10 = completely happy.

V-.2 Main explanatory variables

The main explanatory variables capture socio-economic, demographic and occupational dimensions likely to influence well-being. The variables retained for this study are presented below.

- **Personal income (**revp**)**

"What is your average net monthly personal income (before tax)?"

Coding: 0 = Refused to answer; 1 = Less than €1,500; 2 = €1,500–2,000; 3 = €2,001–2,500; 4 = €2,501–3,000; 5 = €3,001–4,000; 6 = €4,001–6,000; 7 = More than €6,000.

This is an ordinal variable reflecting the individual's income level. Income is one of the most widely studied determinants of subjective well-being. Many studies confirm a robust relationship

between income and happiness, although this relationship is complex. A study covering 166 countries finds an inverted U-shaped relationship between income and happiness in developing countries (Behera, Dash & Dandapat, 2024).

- **Education level (**educ**)**

"What is the highest degree you have obtained?"

Coding: 0 = No degree; 1 = Lower secondary certificate (Brevet/BEPC); 2 = Vocational certificate (CAP/BEP); 3 = Baccalauréat; 4 = Bac+2; 5 = Bac+3/4; 6 = Bac+5; 7 = Bac+7 or higher. This is an increasing ordinal variable: the higher the value, the higher the level of education. Education influences well-being, and empirical results point to a positive relationship. A study from Pakistan (Salameh et al., 2022) shows that education has a positive effect on happiness and life satisfaction, alongside employment and freedom of choice.

- **Sex (**sexe**)**

"What is your sex?"

Coding: 1 = Female; 2 = Male.

For the analysis, the variable was recoded as a binary variable:

• 0 = Female

• 1 = Male

The literature identifies marked gender differences in well-being. Maharlouei et al. (2020) examine the determinants of subjective health and happiness in the United States. Their results show that education, employment and marriage are positively associated with well-being. However, these relationships vary by gender: full-time employment matters more for men's happiness, while education and marriage are more strongly linked to women's happiness.

- **Age (**age**)**

"What is your age group?"

Coding: 1 = 18–29; 2 = 30–39; 3 = 40–49; 4 = 50–59; 5 = 60–69; 6 = 70 and over. This categorical ordinal variable is used to capture generational effects on well-being.

The literature traditionally documents a U-shaped relationship between age and happiness:

• Youth (18–30): high levels of happiness

• Middle age (40–50): low point (nadir) around 48–50

• Older age (60+): happiness rises again

This pattern is explained by:

- Work and family pressures in midlife
- Improved emotional regulation with age
- Refocusing on social relationships and gratitude

For example, Blanchflower (2020) shows in a study of 145 countries that this curve is robust, with the low point at around age 48.

However, recent work points to a change in this pattern: since around 2017, in several developed countries, the well-being of young adults (18–25) has declined sharply, to the point that the classic U-shape is disappearing, with the youngest age group sometimes becoming the least happy (Blanchflower & Bryson, 2025).

- **Smoking (**cig**)**

This variable measures individuals' smoking behavior.

The original responses were coded as follows:

• 0 = No

• 1 = Yes, regular cigarettes

• 2 = Yes, e-cigarettes

To simplify the analysis, the variable was recoded as binary:

• 1 = the individual smokes (in any form)

• 0 = the individual does not smoke

Weinhold and Chaloupka (2017) show that this variable affects well-being. In addition, Piper et al. (2012) followed individuals for three years after an attempt to quit smoking and found that those who successfully quit reported a significant improvement in life satisfaction.

- **Alcohol consumption (**Consmalc**)**

"Over the past 12 months, how often have you consumed alcoholic beverages?"

Coding: 1 = Never; 2 = At least once a month; 3 = At least once a week; 4 = Every day.

Among Norwegian students, Sæther et al. (2019) find that hazardous drinking is associated with slightly lower life satisfaction, but also with more close friends and less loneliness. This suggests that moderate consumption may foster sociability, while excessive consumption harms well-being.

- **Recent financial difficulty (**Difcfinance**)**

"Over the past 12 months, has your household, due to money problems, been unable to pay on time for rent, a mortgage, consumer credit, or utility bills (water, gas, electricity, etc.)?"

Coding: 0 = No, 1 = Yes

This binary variable measures the financial situation experienced over the past 12 months. It reflects the presence of immediate economic constraints or budgetary stress within the household. Using a longitudinal study of young American adults, Rhodes et al. (2024) show that pressure from debt collection agencies (calls, harassment, threats) is associated with increased psychological distress, especially among low-income individuals. These results suggest that it is not only the level of debt, but also the social and institutional interactions surrounding debt collection, that damage mental well-being.

- **Region of residence (**region**)**

"In which region is your main residence located?"

Coding: 1 = Provence-Alpes-Côte d'Azur; 2 = Centre-Val de Loire; 3 = Occitanie; 4 = Nouvelle-Aquitaine; 5 = Hauts-de-France; 6 = Bretagne; 7 = Auvergne-Rhône-Alpes; 8 = Île-de-France; 9 = Pays de la Loire; 10 = Normandie; 11 = Grand Est; 12 = Bourgogne-Franche-Comté; 13 = Corse.

This qualitative variable captures geographical and cultural differences in health behaviors. Regional differences in well-being can be explained, for example, by local economic development (job opportunities and public services). Social capital plays a significant mediating role: an increase in social capital can strengthen social support and thereby improve well-being. Zhou et al. (2022) show that social capital affects mental health through social integration, which strengthens the sense of belonging and support.

- **Dependent children (**ChargEnf**)**

"Do you have dependent children?"

Coding: 0 = No; 1 = Yes. This binary variable represents family responsibilities that may influence well-being.

Having children has an ambivalent effect on subjective well-being. Some studies show that parenthood increases the sense of meaning and life satisfaction, especially when economic resources and social support are sufficient (Nelson et al., 2014). Conversely, financial stress, the burden of care and a lack of family support can reduce parents' happiness, particularly among young households and single parents.

VI- Descriptive Statistics

Distribution of well-being

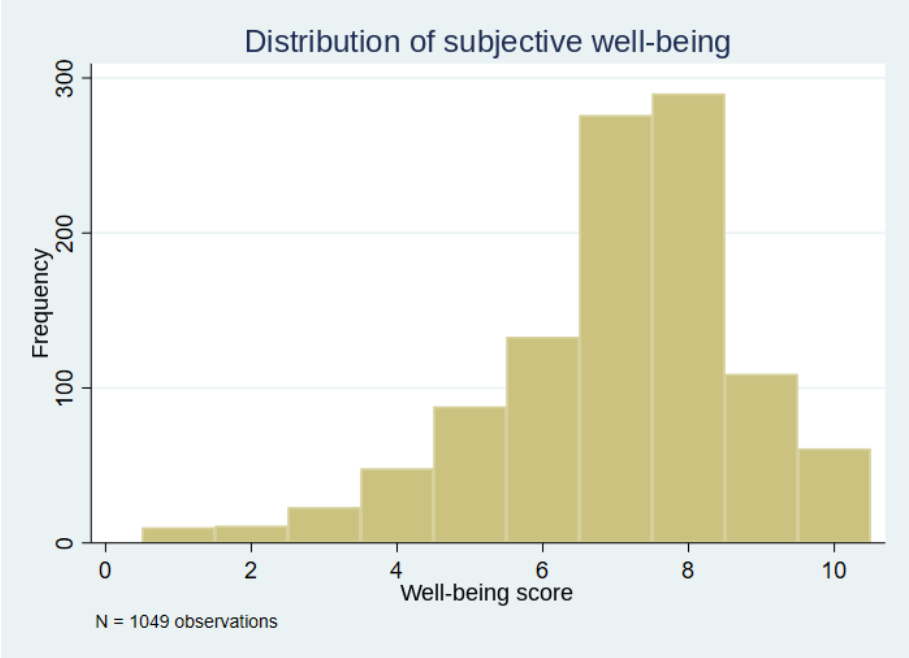


FIGURE 1 : Distribution of well-being

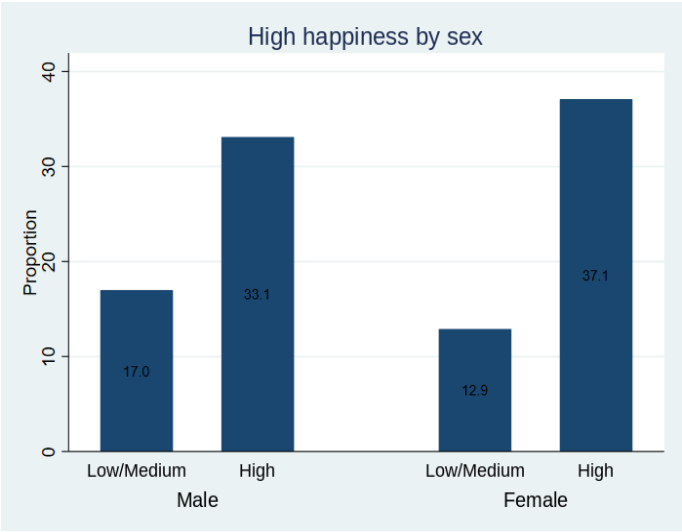
The distribution is heavily concentrated at high values (7–8). It is skewed, indicating that most respondents describe themselves as relatively happy.

Women report high happiness more often (74.2%) than men (66.1%). The chi-squared test confirms that the difference is significant ($p = 0.004$). Overall, 70.2% of the sample reports high happiness.

- Happiness and smoking

The results are as follows:

- Non-smokers: 72.1% report high happiness
- Smokers: 64.9% report high happiness



Key
frequency *row percentage*

sex	High happiness (≥7/10) Low/Med	High	Total
Male	178 33.90	347 66.10	525 100.00
Female	135 25.76	389 74.24	524 100.00
Total	313 29.84	736 70.16	1,049 100.00

Pearson chi2(1) = 8.3031 Pr = 0.004

FIGURE 2 : Well-being by sex

Key
frequency *row percentage*

cig	High happiness (≥7/10) Low/Med	High	Total
Non-smoker	216 27.94	557 72.06	773 100.00
Smoker	97 35.14	179 64.86	276 100.00
Total	313 29.84	736 70.16	1,049 100.00

Pearson chi2(1) = 5.0388 Pr = 0.025

The difference is significant (chi-squared $p = 0.025$), meaning that smokers and non-smokers do not have the same levels of well-being. Non-smokers account for 73.69% of our sample.

- **Happiness and financial difficulties**

Key
frequency *row percentage*

difficulties	High happiness (≥7/10) Low/Med	High	Total
No	247 26.82	674 73.18	921 100.00
Yes	66 51.56	62 48.44	128 100.00
Total	313 29.84	736 70.16	1,049 100.00

Pearson chi2(1) = 32.8667 Pr = 0.000

These statistics show that:

- Without difficulties: 73.2% report high happiness
- With difficulties: 48.4% report high happiness
- The difference is highly significant (chi-squared $p < 0.001$). The well-being of individuals facing financial difficulties is therefore markedly different from that of individuals without such difficulties.

VII- Econometric Methodology

VII-.1 Binary choice models

To analyze the probability of reporting a high level of happiness, we estimate two models:

Logit model

$$P(Y_i = 1 \mid X_i) = \Lambda(X_i'\beta) = \frac{e^{X_i'\beta}}{1 + e^{X_i'\beta}}$$

Probit model

$$P(Y_i = 1 \mid X_i) = \Phi(X_i'\beta) = \int_{-\infty}^{X_i'\beta} \phi(t)\, dt$$

where $Y_i = 1$ if happiness is high, X_i is the vector of explanatory variables, and β is the vector of coefficients to be estimated.

VII-.2 Ordered logit model

To exploit the full ordinal information on happiness levels:

$$P(Y_i = j) = F(\kappa_j - X_i'\beta) - F(\kappa_{j-1} - X_i'\beta)$$

where $j \in \{1, 2, 3\}$ denotes the three happiness levels, F is the logistic cumulative distribution function, and κ_j are the thresholds to be estimated.

VII-.3 Model specification

The full model is written as:

$$\begin{aligned}\text{Happiness}_i = \beta_0 &+ \beta_1 \text{Female}_i + \beta_2 \text{Smoker}_i + \sum_{k=1}^{7} \gamma_k \text{Educ}_{ik} + \sum_{m=2}^{7} \delta_m \text{Income}_{im} \\ &+ \sum_{r=2}^{13} \theta_r \text{Region}_{ir} + \sum_{a=2}^{6} \lambda_a \text{Age}_{ia} + \beta_3 \text{Children}_i \\ &+ \sum_{c=2}^{4} \mu_c \text{Alcohol}_{ic} + \beta_4 \text{FinDiff}_i + \varepsilon_i\end{aligned}$$

Standard errors are robust to heteroskedasticity (Huber-White estimator).

VIII- Results

VIII-.1 Binary logit model

For this regression, the dependent variable bienetre is recoded into two categories: bonheur_eleve = 1 if bienetre >= 7, and 0 otherwise.

The model is globally significant ($\chi^2 = 81.47$, $p < 0.001$), with a pseudo-R^2 of 6.96%. The share of correctly classified observations is 72.9% (**Appendix 1**).

TABLE 1 – Marginal effects from the binary logit model

Variable	Coefficient	Marginal effect	Significance
Female	0.284*	+5.5 points	$p = 0.070$
Smoker	-0.251	-4.9 points	$p = 0.143$
Income (ref.: <€1,500)			
€1,500–2,000	0.544***	+11.2 points	$p = 0.004$
€2,001–2,500	0.606***	+12.3 points	$p = 0.006$
€2,501–3,000	0.685**	+13.7 points	$p = 0.011$
>€6,000	1.917*	+29.2 points	$p = 0.070$
Dependent children	0.600***	+11.0 points	$p = 0.002$
Financial difficulties	-1.170***	-25.5 points	$p < 0.001$

*Significance: *$p < 0,10$, **$p < 0,05$, ***$p < 0,01$*

VIII-.2 Interpretation of marginal effects

The most notable marginal effects are:

- **Financial difficulties**: Having experienced payment difficulties in the past 12 months reduces the probability of reporting high happiness by 25.5 percentage points, all else being equal. This is the strongest effect identified in our model, highlighting the major impact of financial stress on subjective well-being, over and above the level of income.
- **High income** (>€6,000): Compared with individuals earning less than €1,500 per month, those whose income exceeds €6,000 are 29.2 percentage points more likely to report high happiness, all else being equal.

  The income effect is gradual:

  * €1,500–2,000: +11.2 points ($p < 0.01$)
  * €2,001–2,500: +12.3 points ($p < 0.01$)
  * €2,501–3,000: +13.7 points ($p < 0.05$)

  These results do not yet show diminishing marginal returns; on the contrary, the marginal effect appears to increase slightly over this income range.
- **Dependent children**: Having dependent children increases the probability of reporting high happiness by 11.0 percentage points, all else being equal ($p < 0.01$). This result contrasts with some international studies that find negative effects of parenthood on well-

being. It may reflect the effectiveness of French family policies, such as family allowances, childcare provision and parental leave, which ease the stress associated with raising children.

- **Gender**: Women are 5.5 percentage points more likely than men to report high happiness, all else being equal. This effect is marginally significant ($p = 0.070$) and modest in size, but consistent with part of the literature on gender differences in reported subjective well-being.

After estimating the model, we find that education, age, alcohol consumption and region have no significant effect on happiness.

VIII-.3 Interactions: Sex × Financial difficulties

- Men without difficulties: 71.8% report high happiness
- Men with difficulties: 44.1% (–27.7 points)
- Women without difficulties: 77.1% report high happiness
- Women with difficulties: 51.2% (–25.9 points)

Although the interaction is not statistically significant ($p = 0.626$), women maintain a slightly higher level of happiness even when facing difficulties **(Appendix 2)**.

VIII-.4 Logit vs. Probit comparison

TABLE 2 – Comparison of the Logit and Probit models

Criterion	**Logit**	**Probit**
Log-likelihood	-594.87	-595.15
AIC	1265.73	1266.30
BIC	1454.04	1454.61
Pseudo-R^2	0.0696	0.0691

The two models produce virtually identical results. The Logit model is slightly preferred according to the AIC and BIC criteria, but the difference is negligible. Marginal effects are practically identical (difference < 0.3 percentage points).

VIII-.5 Ordered logit model

The ordered model allows us to analyze transitions between three levels of happiness. We define bonheur_cat = 1 if bienetre <= 4, bonheur_cat = 2 if 5 <= bienetre <= 7, and bonheur_cat = 3 if bienetre >= 8.

To check the robustness of the standard ordered logit model, we estimated a generalized ordered logit model (gologit2), which relaxes the proportional odds (parallel lines) assumption.

Test of the proportional odds assumption: The autofit algorithm tests each variable sequentially. Of the 37 variables included in the model:

- 35 variables (95%) satisfy the proportional odds assumption ($p > 0.05$)
- 2 variables violate this assumption:
  * Education level Bac+3/4 ($p = 0.031$)
  * Corse region ($p < 0.001$)

The global test of proportionality gives $\chi^2 = 26.38$ (35 df), $p = 0.853$.

Interpretation : the proportional odds assumption is broadly supported by the global test, although two variable-specific violations remain. These violations concern :

- **Corse region (region 13)**: the effect differs markedly between the two transitions. In the generalized model:
  * Transition Low → Medium: coefficient = +10.99 (strongly positive)
  * Transition Medium → High: coefficient = +0.33 (close to zero)

  This suggests that living in Corse strongly reduces the probability of being in the *Low* category, without significantly affecting the transition to *High*. However, the number of respondents from Corse is very small ($n \approx 3$), which makes this estimate unreliable.
- **Education Bac+3/4**: the difference between transitions is smaller, and the effect remains non-significant in both cases.

Conclusion The standard ologit model with the proportionality constraint is validated and can be used with confidence. The gologit2 model does not substantially change conclusions, except for two marginal variables (Corse and one education level) that are not significant in our main interpretation.

TABLE 3 : Comparison of the ologit and gologit2 models

Criterion	**ologit**	**gologit2**
Log-likelihood	- 931.46	- 928.69
AIC	1940.92	1935.38
BIC	2134.19	2138.83
Number of parameters	39	41

Model comparison: The gologit2 model **(Appendix 3)** slightly improves fit (lower AIC), but the difference is minimal. It is, however, more complex (2 additional parameters), which justifies retaining the standard ologit model for the main interpretation.

Based on the ologit model, the marginal effects are as follows:

TABLE 4 : Marginal effects from the ordered logit model (percentage points)

Variable	**Low (1–4)**	**Medium (5–7)**	**High (8–10)**
Health behaviors			
Smoker (ref.: Non-smoker)	+2.5 pts*	+4.4 pts**	–6.8 pts**
Economic situation			
Income €1,500–2,000	–5.1 pts***	–9.0 pts***	+14.1 pts***
Income €2,001–2,500	–3.4 pts**	–5.1 pts*	+8.5 pts**
Income €2,501–3,000	–4.4 pts**	–7.3 pts**	+11.7 pts**
Income €3,001–4,000	–6.7 pts***	–13.9 pts***	+20.6 pts***
Income >€6,000	–9.2 pts***	–25.8 pts**	+35.0 pts***
Financial difficulties	+8.4 pts***	+10.1 pts***	–18.5 pts***
Family situation			
Dependent children	–3.2 pts***	–6.8 pts***	+10.0 pts***
Region (ref.: PACA)			
Auvergne-Rhône-Alpes	+4.6 pts**	+9.6 pts**	–14.2 pts**
Île-de-France	+3.7 pts*	+8.1 pts*	–11.8 pts*

*Significance: *p < 0,10, **p < 0,05, ***p < 0,01*

VIII-.6 Detailed interpretation of marginal effects

1. **Smoking: a shift toward lower categories** Compared with non-smokers, and all else being equal, being a smoker:

- Increases the probability of low well-being by 2.45 percentage points ($p < 0.10$)
- Increases the probability of medium well-being by 4.37 percentage points ($p < 0.05$)
- Reduces the probability of high well-being by 6.82 percentage points ($p < 0.05$)

Algebraic sum: $2.45 + 4.37 - 6.82 = 0$

Interpretation: Smoking is significantly associated with lower well-being in the ordered model, whereas it was not significant in the binary model ($p = 0.143$). This shows that the effect of smoking operates mainly through a shift toward intermediate happiness categories rather than through a simple binary switch.

Smokers are 6.82 percentage points less likely to reach high happiness, with this probability shifting mainly to the medium category (+4.37 percentage points). This result is consistent with the literature on smoking and well-being (Weinhold & Chaloupka, 2017) and suggests that smoking is not necessarily associated with deep distress, but rather with suboptimal well-being.

Possible mechanisms:

- Nicotine dependence generating chronic stress
- The financial cost of tobacco reducing available resources
- Social stigma attached to smokers
- Effects on physical health (breathing, fatigue)
- Reverse causality: poor well-being may lead to smoking as a *coping* strategy

2. **Income: a gradual and non-linear effect**

The effect of income on happiness follows a gradual pattern, particularly pronounced at the highest income levels.

Income €1,500 - 2,000 (compared with < €1,500):

- Reduces the probability of low well-being by 5.12 percentage points ($p < 0.01$)
- Reduces the probability of medium well-being by 8.99 percentage points ($p < 0.01$)
- Increases the probability of high well-being by 14.11 percentage points ($p < 0.01$)

Interpretation: Even a modest increase in income (moving above the €1,500 threshold) has a substantial and highly significant effect on happiness. Most of the effect comes from a shift out of the medium category (–8.99 percentage points) into the high category (+14.11 percentage

points), indicating that crossing this threshold allows individuals to move beyond satisfactory well-being to high happiness.

Income > €6,000: the largest effect

- Reduces the probability of low well-being by 9.22 percentage points ($p < 0.01$)
- Reduces the probability of medium well-being by 25.80 percentage points ($p < 0.01$)
- Increases the probability of high well-being by 35.02 percentage points ($p < 0.01$)

Interpretation: Earning more than €6,000 per month, compared with less than €1,500, increases the probability of high happiness by more than one third. This large effect comes mainly from a shift out of the medium category (–25.80 points).

This result qualifies the *Easterlin paradox*, according to which the effect of income on happiness levels off. In our French sample, very high incomes continue to have a substantial effect on the probability of reaching the highest levels of happiness.

Possible explanations:

- Complete financial security: no money-related anxiety
- Access to high-quality experiences: travel, leisure, culture
- Control and autonomy: the ability to shape one's life according to one's preferences
- Social status: recognition and esteem associated with high income
- Selection effect: personality traits correlated with happiness (optimism, ambition)

Progression of the income effect:

Income level	Effect on "High"	Cumulative effect
<€1,500 (ref.)	0	0
€1,500 - 2,000	+14.11	+14.11
€2,001 - 2,500	+8.46	+22.57
€2,501 - 3,000	+11.68	+34.25
€3,001 - 4,000	+20.63	+54.88
>€6,000	+35.02	+89.90

Key observation: The estimated effects do not increase monotonically across income categories. The largest positive effect is observed for individuals earning more than €6,000 per month, compared with the reference category. These results suggest a positive association between income and the probability of reporting high well-being, although the magnitude varies across income groups.

3. **Financial difficulties:** Having experienced payment difficulties in the past 12 months, all else being equal:
    - Increases the probability of low well-being by 8.41 percentage points ($p < 0.01$)
    - Increases the probability of medium well-being by 10.07 percentage points ($p < 0.01$)
    - Reduces the probability of high well-being by 18.49 percentage points ($p < 0.01$)

Interpretation: Financial difficulties have a large and highly significant negative effect on happiness. Their effect is comparable in magnitude (but opposite in sign) to that of moving into the €3,001–4,000 income bracket (+20.63 points).

A key distinction from income level: This effect persists after controlling for income. It is therefore not simply "having little money" that reduces happiness, but the concrete experience of financial stress: unpaid bills, pressure from creditors, service disconnections and daily anxiety.

Explanatory mechanisms:

- Acute psychological stress: constant anxiety, loss of control (Rhodes et al., 2024)
- Social stigma: shame and isolation
- Cascading problems: penalties, deteriorating housing conditions
- Radical uncertainty: inability to plan ahead
- Family tensions: conflicts over money

Distribution of the effect: Unlike high income, which mainly shifts individuals from *Medium* to *High*, financial difficulties push individuals more evenly toward *Low* (+8.41) and *Medium* (+10.07), affecting the entire distribution of happiness.

Policy implication: This result underlines the importance of policies to prevent over-indebtedness and support households in difficulty (payment moratoria, banking mediation, social microcredit). The well-being gains from preventing financial difficulties are potentially larger than those from marginal increases in income.

4. **Dependent children: a net positive effect on high happiness**

Compared with people without children, and all else being equal, having dependent children:

- Reduces the probability of being in the *Low* category by 3.20 points ($p < 0.01$)
- Reduces the probability of being in the *Medium* category by 6.77 points ($p < 0.01$)
- Increases the probability of being in the *High* category by 9.98 points ($p < 0.01$)

Interpretation: Parenthood is associated with a significantly higher probability (almost 10 points) of belonging to the highest happiness category. The effect comes mainly from a shift out of the *Medium* category, suggesting that children raise well-being beyond the "satisfactory" level.

Context and originality of the result: This result contrasts with part of the international literature, particularly from the United States (Herbst & Ifcher, 2016), which often finds negative or null effects of parenthood. Several factors may explain this difference in the French context:

Generous family policies:

- Substantial family allowances
- Accessible and subsidized childcare
- Paid parental leave
- The family quotient in income taxation

Other contextual factors:

- Pro-natalist cultural norms
- Work-family balance (35-hour week, right to disconnect)
- Positive selection: parents with dependent children are often those able to take on parenthood
- Income and financial difficulties are controlled by the model

Mechanisms of parental happiness:

- Meaning and fulfillment
- Richer social relationships
- Unconditional love
- A sense of the future

Limitation: The positive effect does not mean that all parents are happier. There is strong heterogeneity: single parents, parents with disabilities or parents in precarious situations may experience much lower well-being. The result reflects an average effect on a largely advantaged population.

5. **Auvergne-Rhône-Alpes region: a significant geographical effect** Compared with Provence-Alpes-Côte d'Azur (reference), and all else being equal, living in Auvergne-Rhône-Alpes:
   - Increases the probability of being in the *Low* category by 4.63 points ($p < 0.05$)
   - Increases the probability of being in the *Medium* category by 9.55 points ($p < 0.05$)
   - Reduces the probability of being in the *High* category by 14.18 points ($p = 0.036$)

Interpretation: Among the 13 regions analyzed, Auvergne-Rhône-Alpes is the only region with an individually statistically significant coefficient for the probability of reporting high well-being, after controlling for socio-economic factors. Living in this region reduces the probability of reporting high happiness by 14 percentage points relative to PACA, an effect comparable in size (though opposite in sign) to that of earning €1,500–2,000 rather than less than €1,500.

Possible explanatory mechanisms:

* *The PACA effect ("sunshine effect")*:
  - A Mediterranean climate conducive to well-being
  - An attractive living environment (sea, natural areas)
  - A positive image of the region (Côte d'Azur, quality of life)
  - Possible selection bias: migration to PACA of people seeking a better quality of life
* *Urban stress in Auvergne-Rhône-Alpes*:
  - Lyon: high cost of living, congestion, pollution
  - Grenoble: constrained topography, air pollution
  - Long commuting times in peri-urban areas
* *Upward social comparison*:
  - Lyon attracts ambitious profiles
  - Potentially more unfavorable social comparisons
* *Differences in social capital*:
  - PACA: denser social networks (Mediterranean culture of social ties)
  - Auvergne-Rhône-Alpes: individualistic urban areas and depopulated rural areas
* *Climatic and environmental factors*:
  - Harsh winters, less sunshine
  - Higher risk of seasonal depression

Île-de-France: a similar but marginally non-significant effect

The Île-de-France region shows a similar effect (–11.79 points for the *High* category), but with $p = 0.074$, just above the 5% threshold. This suggests a consistent pattern: large metropolitan areas, despite their economic opportunities, may be associated with lower subjective well-being.

Limitations:

- Omitted regional variables: air quality, access to green spaces, local public services, community networks
- Migration selection bias: unhappy people may leave certain regions
- Limited sample: small numbers of respondents in some regions

- No significant joint test: the global test of regional effects is not significant ($p = 0.454$), which calls for caution

Implication: Although some individual regional coefficients are statistically significant, the joint test does not provide sufficient evidence of an overall regional effect (p = 0.454). These findings should therefore be interpreted with caution. Further research incorporating environmental and regional characteristics could help better understand geographical differences in subjective well-being.

IX- Robustness Checks

IX-.1 Specification test (Linktest)

The Linktest is a model specification test that detects whether a model is correctly specified. If the model is correctly specified, no additional variable should be able to improve the prediction of the dependent variable. The link specification test reveals no major problem. The squared term (_hatsq) is only marginally significant ($p = 0.052$), suggesting that the functional form is broadly adequate **(Appendix 4)**.

IX-.2 Multicollinearity

The variance inflation factor (VIF) analysis shows:

- Mean VIF: 3.33
- Maximum VIF: 16.54 (for education level 3)
- Most VIFs are below 3, indicating acceptable multicollinearity

The moderate correlation between education and income ($r = 0.34$) is expected and does not compromise the estimation of the coefficients.

X- Additional Analyses (Subgroup Analyses)

Without financial difficulties ($N = 921$):

- Income effect: significant and positive
- Effect of children: significant ($\beta = 0.45$, $p = 0.035$)
- Pseudo-R^2: 0.041

With financial difficulties ($N = 128$):

- Stronger gender effect ($\beta = 1.00$, $p = 0.047$) **(Appendix 6)**

- Income effect: not significant (insufficient sample size)
- Effect of children: $\beta = 0.95$, $p = 0.087$

These results suggest that the mechanisms underlying happiness differ according to financial situation, although the small size of the subsample with difficulties limits the scope of these conclusions.

X-.1 Distribution of predicted probabilities

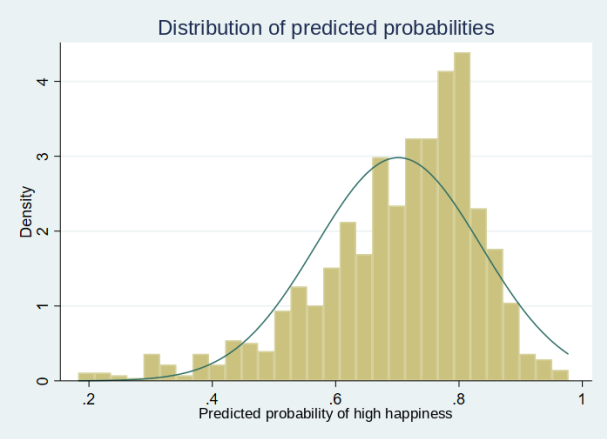


FIGURE 3 : Distribution of predicted probabilities

The predicted probabilities are fairly well dispersed (min = 0.18, max = 0.98), with a concentration around 0.70, which matches the observed share of high happiness. This indicates that the model captures individual variability well.

XI- Conclusion

XI-.1 Main findings

This econometric study, based on 1,049 observations, identifies several significant determinants of happiness. The following marginal effects are drawn from the binary logit model, which estimates the probability of reporting a high level of happiness.

- **Financial difficulties**: the most detrimental factor (−25.5 percentage points), independently of income level. This result is consistent with Rhodes et al. (2024) on financial stress.
- **Income**: a positive and increasing effect, particularly marked at the highest levels (+29.2 points for >€6,000).
- **Dependent children**: a significant increase in happiness (+11.0 points), suggesting a mitigating role of family policies in France.
- **Gender**: a modest but positive effect for women (+5.5 points), consistent with recent literature.
- **Smoking**: associated with lower happiness; the effect is not significant in the binary model but clearer in the ordered model.
- **Education**: no significant direct effect after controlling for income, consistent with an instrumental role.
- **Age and region**: no statistically significant effect was identified. Regional effects are not jointly statistically significant, although some individual regional coefficients are significant.

XI-.2 Policy implications

Our results suggest several avenues for action:

- **Tackling financial insecurity**: policies to prevent over-indebtedness and payment arrears could have a major impact on well-being.
- **Supporting families**: maintaining and strengthening support schemes (allowances, childcare, work-family reconciliation).
- **Preventing risk behaviors**: anti-smoking campaigns may indirectly contribute to well-being.
- **Economic stability**: financial security appears to be as important as the absolute level of income.

XI-.3 Limitations and future research

This study has several limitations:

- **Causality**: cross-sectional data do not allow causal relationships to be inferred.
- **Omitted variables**: no measures of physical health, social relationships or personality.

- **Social desirability bias**: reported happiness may be affected by response bias.
- **Representativeness**: the sample is not perfectly representative of the French population.

Future research could:

- Use panel data to analyze dynamics over time.
- Include objective measures of health and social capital.
- Study the mediating mechanisms between education, income and happiness.
- Explore interaction effects (gender × age, income × education).

Taken together, the estimates confirm that income has a positive and significant effect on well-being: the higher the income, the higher the probability of reporting high happiness. Hypothesis H1 is therefore supported.

Financial difficulties emerge as the most negative factor in the model, sharply reducing the probability of reaching high happiness, even at comparable income levels. Hypothesis H2 is thus clearly supported.

Finally, smoking has a negative but more moderate effect on happiness: although its impact is less systematic than that of the other factors, the estimates still show an unfavorable association between smoking and well-being. Hypothesis H3 is therefore partially supported, as the effect is real but less robust than for income and financial difficulties.

This analysis confirms the complexity of the determinants of happiness and the importance of a multidimensional approach. Subjective well-being is not limited to material living standards: financial stability, family ties and health play a fundamental role. From a public health perspective, promoting happiness requires coordinated interventions that combine economic security, social support and the prevention of risk behaviors.

XII- Appendices

Appendix 1: Binary logit regression and classification table

```
. logit bonheur_eleve i.sexe i.cig i.educ i.revp i.region i.age ///
>     i.chargenf i.consmalc i.difcfinance, vce(robust)

Iteration 0:   log pseudolikelihood = -639.34968
Iteration 1:   log pseudolikelihood = -595.76551
Iteration 2:   log pseudolikelihood = -594.87601
Iteration 3:   log pseudolikelihood = -594.86539
Iteration 4:   log pseudolikelihood = -594.86539

Logistic regression                                     Number of obs =  1,049
                                                        Wald chi2(37) =  81.47
                                                        Prob > chi2   = 0.0000
Log pseudolikelihood = -594.86539                       Pseudo R2     = 0.0696
```

bonheur_eleve	Coefficient	Robust std. err.	z	P>|z|	[95% conf.	interval]
sexe						
Female	.2839096	.1567862	1.81	0.070	-.0233856	.5912049
cig						
Smoker	-.2510963	.1714334	-1.46	0.143	-.5870996	.084907
educ						
1	.1399518	.7013842	0.20	0.842	-1.234736	1.514639
2	.4591407	.6377811	0.72	0.472	-.7908872	1.709169
3	.4701286	.6325648	0.74	0.457	-.7696756	1.709933
4	.0599884	.6380445	0.09	0.925	-1.190556	1.310533
5	.3083172	.650283	0.47	0.635	-.9662141	1.582848
6	.5265514	.671991	0.78	0.433	-.7905266	1.84363
7	.6927442	1.024631	0.68	0.499	-1.315495	2.700983
revp						
2	.5439111	.1885283	2.89	0.004	.1744024	.9134198
3	.6063574	.2197264	2.76	0.006	.1757016	1.037013
4	.6854512	.2706102	2.53	0.011	.1550649	1.215837
5	.5679863	.3024288	1.88	0.060	-.0247633	1.160736
6	.3894966	.4450655	0.88	0.381	-.4828158	1.261809
7	1.917236	1.058017	1.81	0.070	-.1564392	3.990912
region						
2	.3531772	.4387878	0.80	0.421	-.5068312	1.213186
3	.1240348	.354596	0.35	0.726	-.5709606	.8190301
4	.2326468	.3538901	0.66	0.511	-.460965	.9262587
5	-.0002549	.3467687	-0.00	0.999	-.679909	.6793993
6	.3781675	.402533	0.94	0.347	-.4107828	1.167118
7	-.30589	.3293422	-0.93	0.353	-.9513889	.339609
8	-.0053647	.323535	-0.02	0.987	-.6394816	.6287522
9	-.1892336	.405355	-0.47	0.641	-.9837147	.6052476
10	.2366084	.4473389	0.53	0.597	-.6401597	1.113376
11	-.0338246	.3698074	-0.09	0.927	-.7586337	.6909845
12	-.2232667	.442413	-0.50	0.614	-1.09038	.6438468
13	-.6999455	1.07944	-0.65	0.517	-2.815608	1.415717
age						
2	.023571	.2568212	0.09	0.927	-.4797893	.5269313
3	-.2316713	.2618115	-0.88	0.376	-.7448124	.2814697
4	.0033313	.2555053	0.01	0.990	-.49745	.5041125
5	.3208711	.2679559	1.20	0.231	-.2043129	.8460551
6	.0568214	.2795464	0.20	0.839	-.4910796	.6047223
chargenf						
With child(ren)	.5996211	.1978967	3.03	0.002	.2117507	.9874914
consmalc						
2	.0196111	.1891324	0.10	0.917	-.3510815	.3903038
3	-.0018914	.1975264	-0.01	0.992	-.3890361	.3852533
4	.1693661	.3503306	0.48	0.629	-.5172693	.8560016
difcfinance						
Yes	-1.169713	.2139841	-5.47	0.000	-1.589114	-.7503117
_cons	-.0499088	.7189371	-0.07	0.945	-1.459	1.359182

```
. estat classification

Logistic model for bonheur_eleve
```

Classified	True D	True ~D	Total
+	709	257	966
-	27	56	83
Total	736	313	1049

```
Classified + if predicted Pr(D) >= .5
True D defined as bonheur_eleve != 0
```

Sensitivity	Pr(+| D)	96.33%
Specificity	Pr(-|~D)	17.89%
Positive predictive value	Pr(D| +)	73.40%
Negative predictive value	Pr(~D| -)	67.47%
False + rate for true ~D	Pr(+|~D)	82.11%
False - rate for true D	Pr(-| D)	3.67%
False + rate for classified +	Pr(~D| +)	26.60%
False - rate for classified -	Pr(D| -)	32.53%
Correctly classified		72.93%

Appendix 2: Interactions: Sex × Financial difficulties

	Margin	Delta-method std. err.	z	P>|z|	[95% conf.	interval]
sexe#difcfinance						
Male#No	.7175226	.0217505	32.99	0.000	.6748924	.7601529
Male#Yes	.4409009	.0530541	8.31	0.000	.3369167	.5448851
Female#No	.7713793	.0201488	38.28	0.000	.7318884	.8108702
Female#Yes	.5115987	.0530905	9.64	0.000	.4075432	.6156541

Appendix 3: Brant test (via gologit2)

```
Generalized Ordered Logit Estimates                   Number of obs =  1,049
                                                      Wald chi2(39) = 477.39
                                                      Prob > chi2   = 0.0000
Log pseudolikelihood = -928.69012                     Pseudo R2     = 0.0469

 ( 1)  [Low (1-4)]2.age - [Medium (5-7)]2.age = 0
```

Appendix 4: Linktest

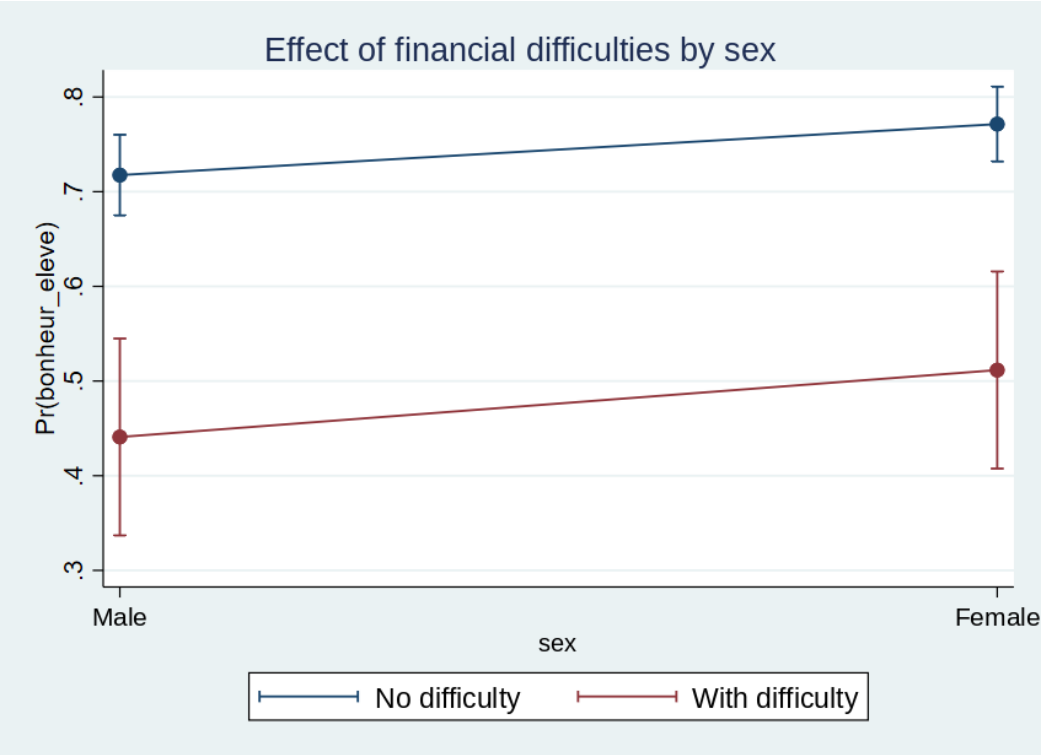
```
. linktest

Iteration 0:   log likelihood = -974.37867
Iteration 1:   log likelihood = -929.54788
Iteration 2:   log likelihood = -929.46954
Iteration 3:   log likelihood = -929.46953

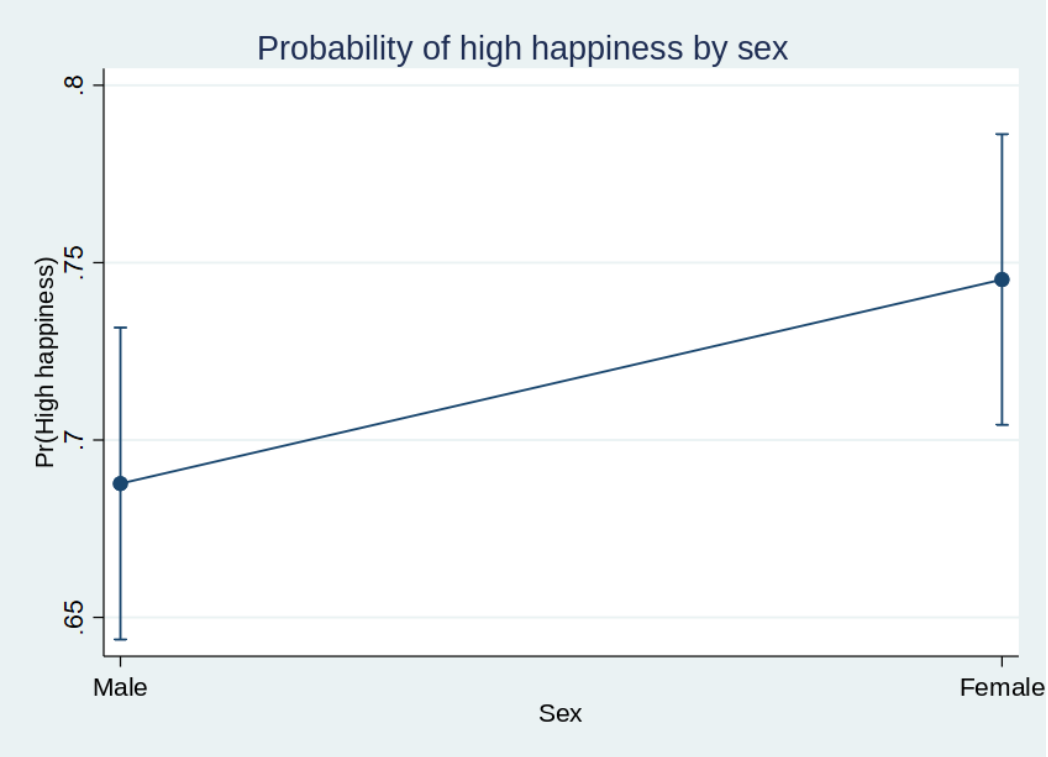
Ordered probit regression                             Number of obs =  1,049
                                                      LR chi2(2)    =  89.82
                                                      Prob > chi2   = 0.0000
Log likelihood = -929.46953                           Pseudo R2     = 0.0461
```

bonheur_cat	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
_hat	1.287925	.1848297	6.97	0.000	.9256658	1.650185
_hatsq	-.4215166	.2170333	-1.94	0.052	-.8468941	.0038609
/cut1	-1.034716	.0660701			-1.164211	-.9052207
/cut2	.5710746	.0602583			.4529706	.6891786

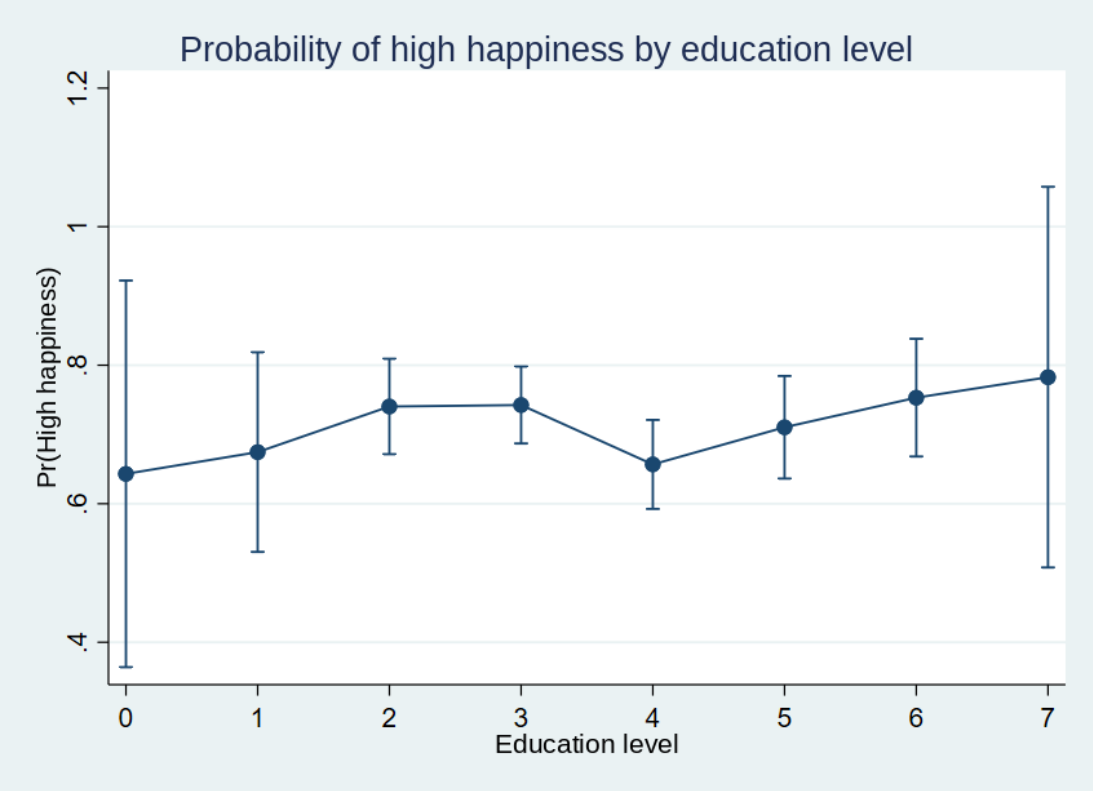
Appendix 5: Effect of financial difficulties by sex



Appendix 6: Probability of high happiness by sex



Appendix 7: Probability of high happiness by education level



XIII- References

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